



A1	2020	☒	☐
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By expressing $\sqrt{3} \sin x + 3 \cos x$ in the form $A \sin(x + \alpha)$, solve

$\sqrt{3} \sin x + 3 \cos x = \sqrt{3}$, for $0 \leq x \leq 2\pi$.

4

$$\begin{aligned}\sqrt{3} \sin x + 3 \cos x &= A \sin(x + \alpha) \\ &= A \sin x \cos \alpha + A \cos x \sin \alpha \\ \therefore A \cos \alpha &= \sqrt{3} \quad (1) \\ A \sin \alpha &= 3 \quad (2) \\ \text{dividing, } \tan \alpha &= \frac{3}{\sqrt{3}} = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}. \\ \therefore A \cos \frac{\pi}{3} &= \sqrt{3} \Rightarrow A = 2\sqrt{3}\end{aligned}$$

$$\begin{aligned}\therefore \sqrt{3} \sin x + 3 \cos x &= 2\sqrt{3} \sin\left(x + \frac{\pi}{3}\right) \\ \text{so } 2\sqrt{3} \sin\left(x + \frac{\pi}{3}\right) &= \sqrt{3} \\ \sin\left(x + \frac{\pi}{3}\right) &= \frac{1}{2} \\ x + \frac{\pi}{3} &= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \\ x &= \frac{\pi}{2}, \frac{11\pi}{6} \text{ given } 0 \leq x \leq 2\pi.\end{aligned}$$

A2	2020	☒	☐
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A device playing a signal given by $x = \sqrt{2} \sin t + \cos t$ produces distortion whenever $|x| \geq 1.5$.

4

For what fraction of the time will the device produce distortion if the signal is played continuously?

$$x = \sqrt{2} \sin t + \cos t = \sqrt{3} \left(\frac{1}{\sqrt{3}} \cos t + \frac{\sqrt{2}}{\sqrt{3}} \sin t \right) = \sqrt{3} \cos(t - \alpha), \text{ where } \tan \alpha = \sqrt{2}.$$

$$\text{Thus } |x| \geq 1.5 \text{ whenever } |\cos(t - \alpha)| \geq \frac{\sqrt{3}}{2}.$$

Now the region in the interval $\left[0, \frac{\pi}{2}\right]$ where $\cos \theta \geq \frac{\sqrt{3}}{2}$ is $\left[0, \frac{\pi}{6}\right]$, and other intervals

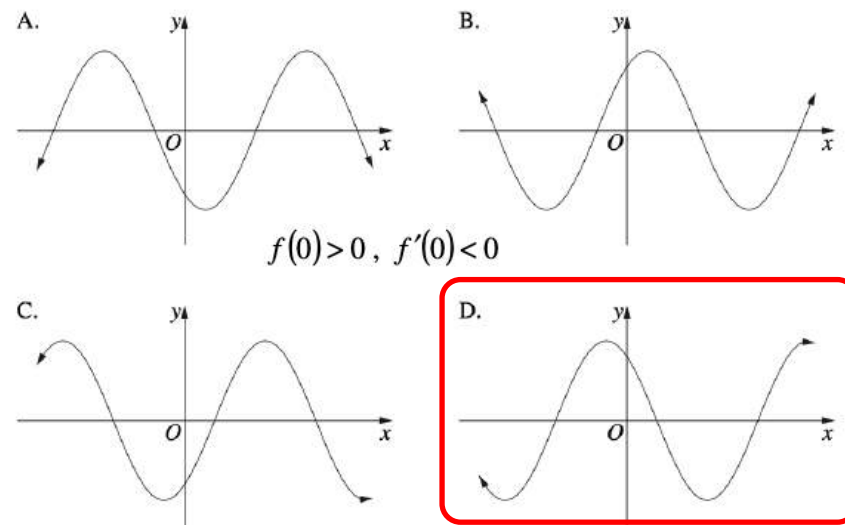
between multiples of $\frac{\pi}{2}$ are similar.

So distortion occurs $\frac{1}{3}$ of the time.

A3	2021	☒	☐
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Which curve best represents the graph of the function $f(x) = -a \sin x + b \cos x$ given that the constants a and b are both positive?

1



A4	2022	☒	☐
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Express $\sqrt{3} \sin(x) - 3 \cos(x)$ in the form $R \sin(x + \alpha)$.

3

$$\text{Q11e } R \sin(x + \alpha) = R \cos \alpha \sin x + R \sin \alpha \cos x$$

$$R \cos \alpha = \sqrt{3}, \quad R \sin \alpha = -3$$

$$\text{Squaring both equations and adding: } R^2 = 12, \quad R = \pm 2\sqrt{3}$$

$$\text{Dividing: } \tan \alpha = -\sqrt{3}, \quad \alpha = -\frac{\pi}{3}, \frac{2\pi}{3}, \dots. \text{ Two possible forms are}$$

$$\sqrt{3} \sin x - 3 \cos x = 2\sqrt{3} \sin\left(x - \frac{\pi}{3}\right) \text{ or } -2\sqrt{3} \sin\left(x + \frac{2\pi}{3}\right)$$



A5	2023	☒	☐
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Solve $\cos \theta + \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.

3

$$\cos \theta + \sin \theta = 1 \quad \theta \in [0, 2\pi]$$

$$\text{Let } \cos \theta + \sin \theta = R \sin(\theta + \alpha) = R \sin \theta \cos \alpha + R \cos \theta \sin \alpha$$

$$\therefore \begin{cases} R \sin \alpha = 1 \\ R \cos \alpha = 1 \end{cases} \quad \therefore \begin{cases} R = \sqrt{1^2 + 1^2} = \sqrt{2} \\ \tan \alpha = 1 \quad \therefore \alpha = \frac{\pi}{4} \end{cases}$$

$$\therefore \sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right) = 1$$

$$\sin\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \quad \theta + \frac{\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \dots \quad \theta = 0, \frac{\pi}{2}, 2\pi$$

A6	2024	☒	☐
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For real numbers a and b , where $a \neq 0$ and $b \neq 0$, we can find numbers $\alpha, \beta, \gamma, \delta$ and R such that $a \cos x + b \sin x$ can be written in the following 4 forms:

1

$$R \sin(x + \alpha) \quad R \sin(x - \beta) \quad R \cos(x + \gamma) \quad R \cos(x - \delta)$$

where $R > 0$ and $0 < \alpha, \beta, \gamma, \delta < 2\pi$.

What is the value of $\alpha + \beta + \gamma + \delta$?

A. 0 B. π C. 2π **D. 4π**

$$\text{Q10 } \sin(x + \alpha) = \sin x \cos \alpha + \cos x \sin \alpha \quad \therefore a = \sin \alpha \text{ and } b = \cos \alpha$$

$$\sin(x - \beta) = \sin x \cos \beta - \cos x \sin \beta \quad \therefore a = -\sin \beta \text{ and } b = \cos \beta$$

$$\cos(x + \gamma) = \cos x \cos \gamma - \sin x \sin \gamma \quad \therefore a = \cos \gamma \text{ and } b = -\sin \gamma$$

$$\cos(x - \delta) = \cos x \cos \delta + \sin x \sin \delta \quad \therefore a = \cos \delta \text{ and } b = \sin \delta$$

$$\therefore a = \cos \gamma = \cos \delta \quad \therefore \delta = 2\pi - \gamma$$

$$\text{and } b = \cos \beta = \cos \alpha \quad \therefore \beta = 2\pi - \alpha \quad \therefore \alpha + \beta + \gamma + \delta = 4\pi$$

D

A7	2019	☒	☐
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A particle is moving along the x -axis in simple harmonic motion. The position of the particle is given by

$$x = \sqrt{2} \cos 3t + \sqrt{6} \sin 3t, \text{ for } t \geq 0.$$

(i) Write x in the form $R \cos(3t - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2

(ii) Find the two values for x where the particle comes to rest. 1

(iii) When is the first time that the speed of the particle is equal to half of its maximum speed? 2

$$\text{Q12bi } x = R \cos(3t - \alpha) = R \cos 3t \cos \alpha + R \sin 3t \sin \alpha \text{ where}$$

$$R \cos \alpha = \sqrt{2} \text{ and } R \sin \alpha = \sqrt{6}, \therefore R = 2\sqrt{2} \text{ and } \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$$

$$\therefore x = 2\sqrt{2} \cos\left(3t - \frac{\pi}{3}\right)$$

$$\text{Q12bii } x = \pm 2\sqrt{2}$$

$$\text{Q12biii } \dot{x} = -6\sqrt{2} \sin\left(3t - \frac{\pi}{3}\right), \text{ maximum speed} = 6\sqrt{2}$$

$$\text{Let } \sin\left(3t - \frac{\pi}{3}\right) = \pm \frac{1}{2}, \quad 3t - \frac{\pi}{3} = -\frac{\pi}{6} \text{ or } \frac{\pi}{6}, \therefore t = \frac{\pi}{18} \text{ or } \frac{\pi}{6}$$

$$t = \frac{\pi}{18} \text{ is the first time.}$$

A8	2018	☒	☐
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Write $\sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2

$$\begin{aligned} \sqrt{3} \sin x + \cos x &= R \sin(x + \alpha), \quad R > 0, \quad 0 \leq \alpha \leq \frac{\pi}{2} \\ &= R \sin x \cos \alpha + R \cos x \sin \alpha \end{aligned}$$

$$\therefore R \cos \alpha = \sqrt{3} \quad \text{①}$$

$$R \sin \alpha = 1 \quad \text{②}$$

$$\text{②} \div \text{①} \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\text{②}^2 + \text{①}^2 \quad R^2(\sin^2 \alpha + \cos^2 \alpha) = 1 + 3$$

$$R^2 = 4$$

$$R = 2 \quad (R > 0)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin\left(x + \frac{\pi}{6}\right)$$



A9	2017	☒	☐
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What is the value of $\tan \alpha$ when the expression $2\sin x - \cos x$ is written in the form $\sqrt{5}\sin(x - \alpha)$? 1

- A. -2 B. $-\frac{1}{2}$ **C. $\frac{1}{2}$** D. 2

Q4 $\sqrt{5}\sin(x - \alpha) = \sqrt{5}\sin x \cos \alpha - \sqrt{5}\cos x \sin \alpha = 2\sin x - \cos x$

$\therefore \sqrt{5}\sin \alpha = 1$ and $\sqrt{5}\cos \alpha = 2$, $\therefore \tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{1}{2}$

A10	2016	☒	☐
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The displacement x of a particle at time t is given by 1

$$x = 5\sin 4t + 12\cos 4t.$$

What is the maximum velocity of the particle?

- (A) 13 (B) 28 **(C) 52** (D) 68

$$\frac{dx}{dt} = 20\sin 4t - 48\cos 4t, \max v = \sqrt{20^2 + 48^2} = 52$$

A11	2015	☒	☐
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Express $5\cos x - 12\sin x$ in the form $A\cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$. 2

$$A\cos(x + \alpha) = A\cos x \cos \alpha - A\sin x \sin \alpha = 5\cos x - 12\sin x$$

$$\therefore A\cos \alpha = 5 \text{ and } A\sin \alpha = 12 \therefore A = 13 \text{ and } \alpha = \tan^{-1}\left(\frac{12}{5}\right)$$

$$\therefore 5\cos x - 12\sin x = 13\cos\left(x + \tan^{-1}\left(\frac{12}{5}\right)\right)$$

A12	2014	☒	☐
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Which expression is equal to $\cos x - \sin x$? 1

- (A) $\sqrt{2}\cos\left(x + \frac{\pi}{4}\right)$** (C) $2\cos\left(x + \frac{\pi}{4}\right)$
(B) $\sqrt{2}\cos\left(x - \frac{\pi}{4}\right)$ (D) $2\cos\left(x - \frac{\pi}{4}\right)$

Q2 $\cos x - \sin x = A\cos(x + b) = A\cos x \cos b - A\sin x \sin b$

$\therefore A\cos b = 1$ and $A\sin b = 1$, $\therefore \tan b = 1$, $b = \frac{\pi}{4}$ and $A = \sqrt{2}$

A13	2013	☒	☐
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(i) Write $\sqrt{3}\cos x - \sin x$ in the form $2\cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. 1

(ii) Hence, or otherwise, solve $\sqrt{3}\cos x = 1 + \sin x$, where $0 < x < 2\pi$. 2

Q12ai

$$\sqrt{3}\cos x - \sin x = 2\cos(x + \alpha) = 2\cos x \cos \alpha - 2\sin x \sin \alpha$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2} \text{ and } \sin \alpha = \frac{1}{2}, \therefore \tan \alpha = \frac{1}{\sqrt{3}}, \alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3}\cos x - \sin x = 2\cos\left(x + \frac{\pi}{6}\right)$$

Q12aii $\sqrt{3}\cos x = 1 + \sin x$ and $0 < x < 2\pi$

$$\therefore \sqrt{3}\cos x - \sin x = 2\cos\left(x + \frac{\pi}{6}\right) = 1, \therefore \cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$

$$\therefore x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \therefore x = \frac{\pi}{6}, \frac{3\pi}{2}$$



A14	2012	☒	☐
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A particle is moving in a straight line according to the equation

$$x = 5 + 6 \cos 2t + 8 \sin 2t,$$

where x is the displacement in metres and t is the time in seconds.

When is the displacement of the particle zero for the first time?

3

Q13cii Let $A \cos(2t - \varepsilon) = 6 \cos 2t + 8 \sin 2t$.

$$\therefore A \cos 2t \cos \varepsilon + A \sin 2t \sin \varepsilon = 6 \cos 2t + 8 \sin 2t$$

$$\therefore A \cos \varepsilon = 6 \text{ and } A \sin \varepsilon = 8$$

$$\therefore A^2 \cos^2 \varepsilon + A^2 \sin^2 \varepsilon = 100, \therefore A = 10$$

$$\frac{\sin \varepsilon}{\cos \varepsilon} = \tan \varepsilon = \frac{8}{6}, \therefore \varepsilon = \tan^{-1} \frac{4}{3}$$

$$\therefore x = 5 + 6 \cos 2t + 8 \sin 2t = 5 + 10 \cos \left(2t - \tan^{-1} \frac{4}{3} \right)$$

$$\text{When } x = 0, 5 + 10 \cos \left(2t - \tan^{-1} \frac{4}{3} \right) = 0,$$

$$\cos \left(2t - \tan^{-1} \frac{4}{3} \right) = -\frac{1}{2}, \therefore 2t - \tan^{-1} \frac{4}{3} = \frac{2\pi}{3},$$

$$t = \frac{1}{2} \left(\frac{2\pi}{3} + \tan^{-1} \frac{4}{3} \right) \approx 1.51 \text{ s}$$

A15	2010	☒	☐
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(i) Express $2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right)$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

3

(ii) Hence, or otherwise, solve $2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 3$, for $0 < \theta < 2\pi$.

2

Q4bi

$$R \cos(\theta + \alpha) = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 2 \cos \theta + 2 \cos \theta \cos \frac{\pi}{3} - 2 \sin \theta \sin \frac{\pi}{3}$$

$$= 3 \cos \theta - \sqrt{3} \sin \theta$$

$$\therefore R \cos \alpha = 3 \text{ and } R \sin \alpha = \sqrt{3}$$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 9 + 3, \therefore R = 2\sqrt{3}$$

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{\sqrt{3}}{3}, \therefore \tan \alpha = \frac{1}{\sqrt{3}}, \alpha = \frac{\pi}{6}$$

$$\text{Hence } 2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 2\sqrt{3} \cos \left(\theta + \frac{\pi}{6} \right)$$

$$\text{Q4bii } 2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 3 \text{ and } 0 < \theta < 2\pi$$

$$2\sqrt{3} \cos \left(\theta + \frac{\pi}{6} \right) = 3, \therefore \cos \left(\theta + \frac{\pi}{6} \right) = \frac{\sqrt{3}}{2}$$

$$\therefore \theta + \frac{\pi}{6} = \frac{11\pi}{6}, \theta = \frac{5\pi}{3}$$



B1	2020	☒	☐
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Given that $\cos \theta - 2 \sin \theta + 2 = 0$, which of the following shows the two possible values for $\tan \frac{\theta}{2}$? 1

- A. -3 or -1 B. -3 or 1 C. -1 or 3 **D. 1 or 3**

B2	2023	☒	☐
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Solve $\cos \theta + \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$. 3

$$\begin{array}{lll}
 \text{Let } t = \tan \frac{\theta}{2} & \therefore t = 0 & \text{or } t = 1 \\
 & \tan \frac{\theta}{2} = 0 & \tan \frac{\theta}{2} = 1 \\
 \therefore \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} = 1 & \frac{\theta}{2} = 0, \pi, \cancel{2\pi}, \dots & \frac{\theta}{2} = \frac{\pi}{4}, \cancel{\frac{5\pi}{4}}, \dots \\
 1-t^2+2t=1+t^2 & \theta = 0, \frac{2\pi}{2} & \theta = \frac{\pi}{2} \\
 2t^2-2t=0 & \therefore \theta = 0, \frac{\pi}{2}, 2\pi & \\
 2t(t-1)=0 & &
 \end{array}$$



C1	2020	☒	☐
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(i) Show that $\sin^3 \theta - \frac{3}{4} \sin \theta + \frac{\sin(3\theta)}{4} = 0$. 2

(ii) By letting $x = 4 \sin \theta$ in the cubic equation $x^3 - 12x + 8 = 0$. 2

Show that $\sin(3\theta) = \frac{1}{2}$.

(iii) Prove that $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$. 2

Q14bi $\frac{\sin 3\theta}{4} = \frac{\sin(2\theta + \theta)}{4} = \frac{\sin 2\theta \cos \theta + \cos 2\theta \sin \theta}{4}$
 $= \frac{2 \sin \theta \cos^2 \theta + (1 - 2 \sin^2 \theta) \sin \theta}{4} = \frac{3}{4} \sin \theta - \sin^3 \theta$
 $\therefore \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{\sin 3\theta}{4} = 0$

Q14biii $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18}$
 $= \frac{1}{2} \left(1 - \cos \frac{\pi}{9} \right) + \frac{1}{2} \left(1 - \cos \frac{5\pi}{9} \right) + \frac{1}{2} \left(1 - \cos \frac{25\pi}{9} \right)$
 $= \frac{1}{2} \left[3 - \left(\cos \frac{\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{25\pi}{9} \right) \right]$
 $= \frac{1}{2} \left[3 - \left(\cos \frac{\pi}{9} + 2 \cos \frac{5\pi}{3} \cos \frac{10\pi}{9} \right) \right]$
 $= \frac{1}{2} \left[3 - \left(\cos \frac{\pi}{9} + \cos \frac{10\pi}{9} \right) \right] = \frac{1}{2} \left[3 - \left(\cos \frac{\pi}{9} - \cos \frac{\pi}{9} \right) \right] = \frac{3}{2}$

Q14bii $x = 4 \sin \theta$ and $x^3 - 12x + 8 = 0$
 $\therefore 64 \sin^3 \theta - 48 \sin \theta + 8 = 0, \sin^3 \theta - \frac{3}{4} \sin \theta + \frac{1}{8} = 0$.

Comparing with bi, $\frac{\sin 3\theta}{4} = \frac{1}{8} \therefore \sin 3\theta = \frac{1}{2}$

C2	2021	☒	☐
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(i) The numbers A , B and C are related by the equations $A = B - d$ and $C = B + d$, where d is a constant. 2

Show that $\frac{\sin A + \sin C}{\cos A + \cos C} = \tan B$.

(ii) Hence, or otherwise, solve $\frac{\sin \frac{5\theta}{7} + \sin \frac{6\theta}{7}}{\cos \frac{5\theta}{7} + \cos \frac{6\theta}{7}} = \sqrt{3}$, for $0 \leq \theta \leq 2\pi$. 2

Q13di

$$\frac{\sin A + \sin C}{\cos A + \cos C} = \frac{\sin(B-d) + \sin(B+d)}{\cos(B-d) + \cos(B+d)} = \frac{2 \sin B \cos d}{2 \cos B \cos d} = \tan B$$

Q13dii Let $B = \frac{11\theta}{14}$ and $d = \frac{\theta}{14}$, $0 \leq \theta \leq 2\pi$

$$\therefore \frac{\sin \frac{5\theta}{7} + \sin \frac{6\theta}{7}}{\cos \frac{5\theta}{7} + \cos \frac{6\theta}{7}} = \tan \frac{11\theta}{14} = \sqrt{3} \therefore \frac{11\theta}{14} = \frac{\pi}{3}, \frac{4\pi}{3} \therefore \theta = \frac{14\pi}{33}, \frac{56\pi}{33}$$

C3	2020	☒	☐
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(i) Prove the trigonometric identity $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. 3

(ii) Hence find expressions for the exact values of the solutions to the equation $8x^3 - 6x = 1$. 4

(i) $\cos 3\theta = \cos \theta \cos 2\theta - \sin \theta \sin 2\theta$ angle sum formula
 $= \cos \theta (\cos^2 \theta - \sin^2 \theta) - \sin \theta (2 \sin \theta \cos \theta)$ angle sum formula
 $= \cos^3 \theta - 3 \cos \theta \sin^2 \theta$
 $= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$ $\cos^2 \theta + \sin^2 \theta = 1$
 $= 4 \cos^3 \theta - 3 \cos \theta$

(ii) Writing $x = \cos \theta$ we get $8 \cos^3 \theta - 6 \cos \theta = 2 \cos 3\theta = 1$.

Consequently $\cos 3\theta = \frac{1}{2}$, and so $3\theta = \pm \frac{\pi}{3} + 2n\pi$, where n is an integer.

Thus $\theta = \pm \frac{\pi}{9} + n \frac{2\pi}{3}$, and so $x = \cos \left(\frac{\pi}{9} \right)$ or $\cos \left(\frac{7\pi}{9} \right)$ or $\cos \left(\frac{13\pi}{9} \right)$.

NB $\cos \left(\frac{7\pi}{9} \right) = -\cos \left(\frac{2\pi}{9} \right)$ and $\cos \left(\frac{13\pi}{9} \right) = \cos \left(\frac{5\pi}{9} \right) = -\cos \left(\frac{4\pi}{9} \right)$.

C4	2021	☒	☐
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By factorising, or otherwise, solve $2 \sin^3 x + 2 \sin^2 x - \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$. 3

Q11g $2 \sin^3 x + 2 \sin^2 x - \sin x - 1 = 0$ and $0 \leq x \leq 2\pi$

$$\sin x (2 \sin^2 x - 1) + (2 \sin^2 x - 1) = 0, (\sin x + 1)(2 \sin^2 x - 1) = 0$$

$$\therefore \sin x = -1, \pm \frac{1}{\sqrt{2}} \therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{3\pi}{2}$$



C5	2020	☒	☐
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Solve $\cos \theta + \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.

3

Q11e $\cos \theta + \sin \theta = 1$ and $0 \leq \theta \leq 2\pi$,

$$2\cos^2 \frac{\theta}{2} - 1 + 2\sin \frac{\theta}{2} \cos \frac{\theta}{2} = 1 \quad \therefore \cos^2 \frac{\theta}{2} - 1 + \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 0,$$

$$\sin \frac{\theta}{2} \cos \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = 0, \quad \sin \frac{\theta}{2} \left(\cos \frac{\theta}{2} - \sin \frac{\theta}{2} \right) = 0$$

$$\therefore \sin \frac{\theta}{2} = 0 \text{ or } \sin \frac{\theta}{2} = \cos \frac{\theta}{2} \text{ where } 0 \leq \theta \leq 2\pi \quad \therefore \theta = 0, \frac{\pi}{2}, 2\pi$$

C6	2016	☒	☐
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At a particular location, the high tide is 9 metres and the low tide is 1 metre.

At this location the tide completes 2 full periods every 25 hours.

Let t be the time in hours after the first high tide today.

(i) Explain why the tide can be modelled by the function $x = 5 + 4\cos\left(\frac{4\pi}{25}t\right)$. 2

(ii) The first high tide tomorrow is at 2 am.

What is the earliest time tomorrow at which the tide is increasing at the fastest rate? 2

Q13ai Amplitude $\frac{9-1}{2} = 4$, centre $= 1 + 4 = 5$, period $T = \frac{25}{2}$

$$\therefore \frac{2\pi}{n} = \frac{25}{2}, \quad n = \frac{4\pi}{25}. \text{ First high tide at } t = 0, \therefore \text{ use the cosine}$$

function to model the tide. Let x metres be the depth of water at the

particular location, $x = 5 + 4\cos\left(\frac{4\pi}{25}t\right)$.

Q13aii Let the first high tide tomorrow is at $t = 0$ (2 am).

$$\text{The tide increases at the fastest rate at } t = \frac{3}{4} \times \frac{25}{2} = \frac{75}{8}$$

i.e. 9 hours 22 min 30 s after 2 am

$\therefore$ The earliest time is 11:22:30 am.